Plastics & their Disposal

Exam code: 0620 & 0971

Download PDF

Written by: Alexandra Brennan
Reviewed by: Stewart Hird
Updated on 10 December 2024

Plastics & their disposal

- Plastics are made from polymers
- Many polymers are chemically unreactive, which means that they are **non-biodegradable**
- This means that the disposal of plastics can cause environmental issues

Incineration

- Polymers release a lot of heat energy when they burn and produce carbon dioxide, which is a greenhouse gas that contributes to climate change
- Some polymers release **toxic fumes** when they burn
  - An example of this is poly(vinylchloride), which releases toxic hydrogen chloride gas when burned
- If incinerated by incomplete combustion, carbon monoxide will be produced which is a toxic gas

Polluting oceans

- Plastic waste is **accumulating in oceans** and causing huge disruptions to marine life

Landfills

- Waste polymers are disposed of in landfill sites
  - But, this takes up valuable land
- Most polymers are non-biodegradable
  - This means that micro-organisms, such as decomposers, cannot break them down
  - This causes sites to quickly fill up

Disposal of polymers is an environmental problem

Help me understand

Stuck on something?

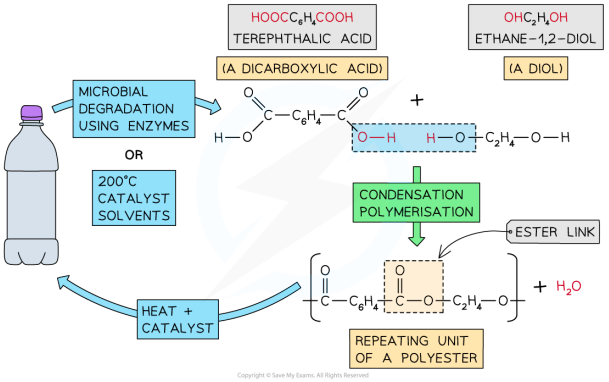
Ask questions and get help with exactly what you're revising.

Got it

Extended tier only

- PET stands for polyethylene terephthalate
  - This is a common polymer used to make things like plastic bottles
- It is a condensation polymer consisting of repeating ester units
  - So, PET is a type of polyester, like terylene
- One of the problems with recycling polymers is that the conditions needed to break them down, which are usually high temperatures and pressures, can degrade the monomers making them unusable for re-polymerisation
- PET is relatively easy to convert back into the monomers
- It can be depolymerised either using enzymes or by chemical methods
- Enzymes present in microbes breakdown the PET into the original monomers
- The same can be achieved using solvents a catalyst and mild heating
- The monomers are recovered and be **re-polymerised** into new PET
- This saves on resources and energy, reducing the carbon footprint of the production process

The re-polymerisation of PET



The breakdown of PET into its two monomers takes place using enzymes or chemical catalysts and mild conditions

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.